



Google Developer Group
On Campus

TechSprint



Leveraging the power of AI



Team Details

- Team name: **HINDCRAFT**
- Team leader name: Pranav Mhasaye
- Problem Statement: Open Innovation - Smart City & Safety

INTRODUCING LUMINOUS

Illuminating the Safest Path, Not Just the Fastest.

Team Name: **HINDCRAFT**



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THE GAP IN MODERN NAVIGATION

Problem (The "Darkness"):

- Existing navigation apps (like Google Maps) optimize for Speed and Distance.
- They often direct vulnerable users (women, students, elderly) through dark, unlit, or deserted alleyways to save 2 minutes.

Solution (The "Light"):

- "Luminous is a safety-first navigation layer built on Google Maps."
- "It calculates a 'Safety Score' for every route based on street lighting, crowd density, and police presence, guiding users to the safest path."

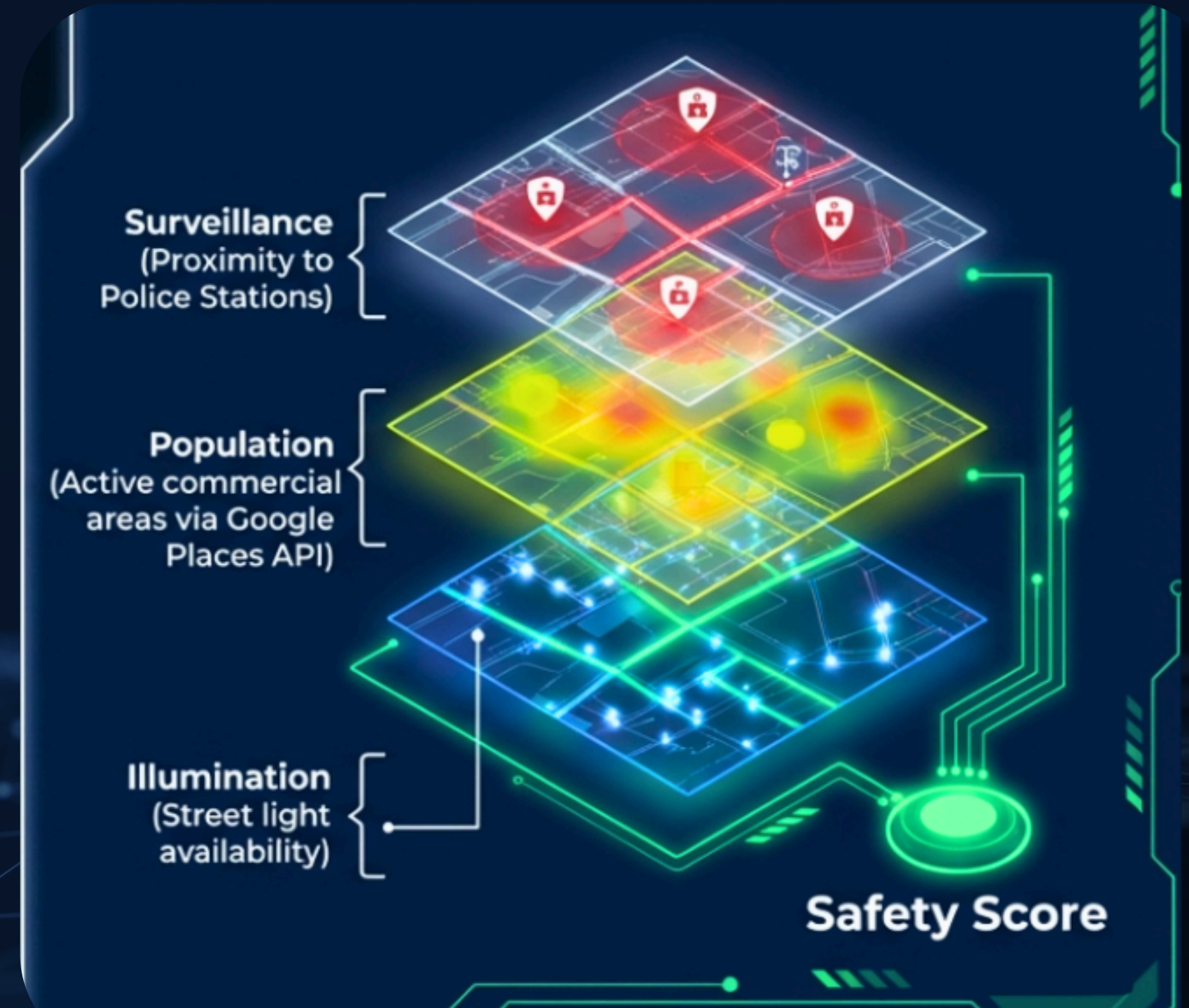
Opportunities

a. How is it different?

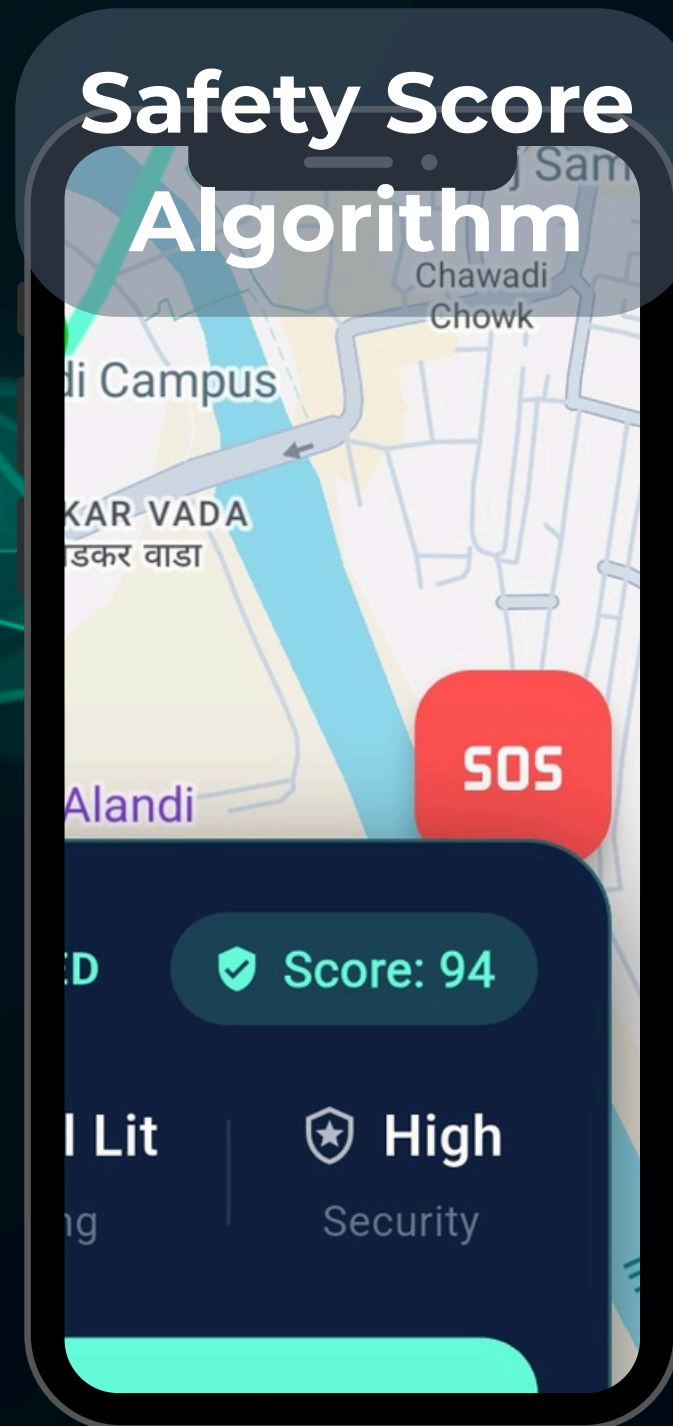
- Standard Maps: "Take this route, it is 5 minutes faster." (Blind to safety).
- Luminous: "Take this route, it is well-lit and has open shops, even if it takes 5 minutes more." (Prioritizes life).

b. How does it solve the problem?

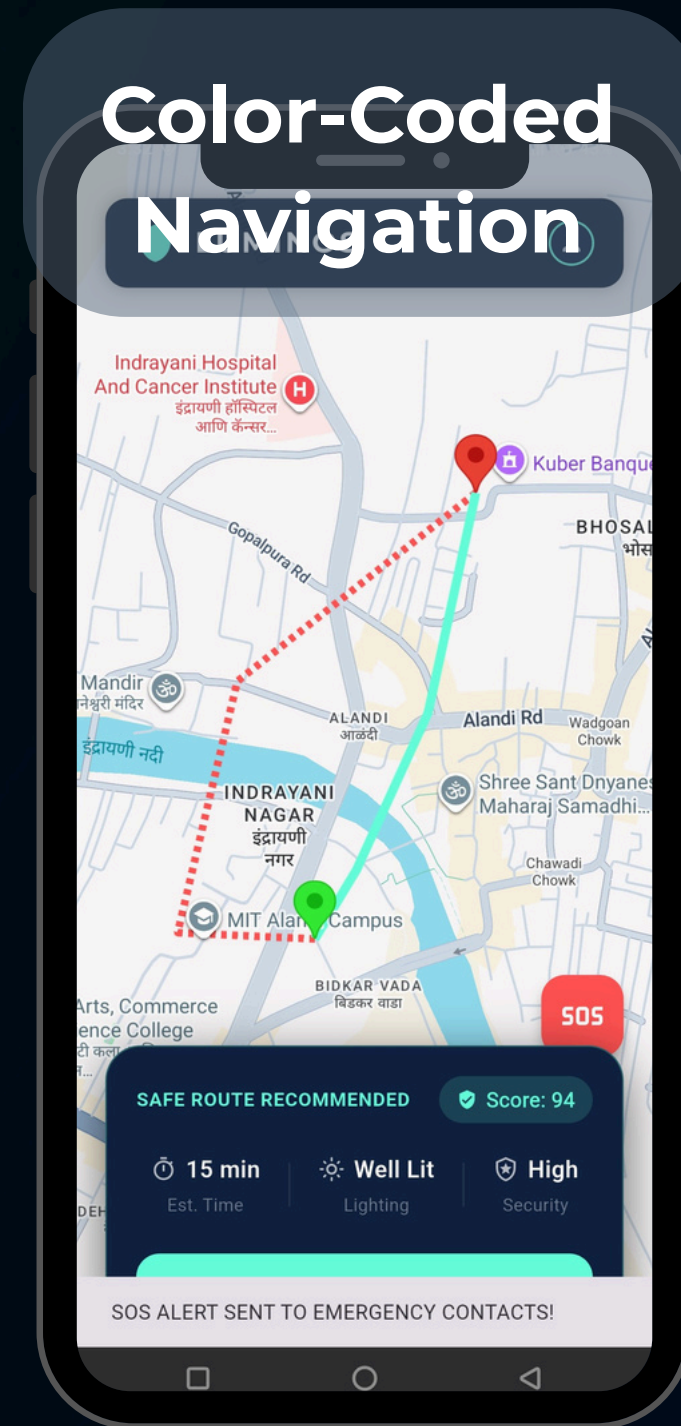
- It aggregates Hyper-local Data (Simulated for Prototype):
- Illumination: Street light availability.
- Population: Active commercial areas (Shops = Safety).
- Surveillance: Proximity to Police Stations.



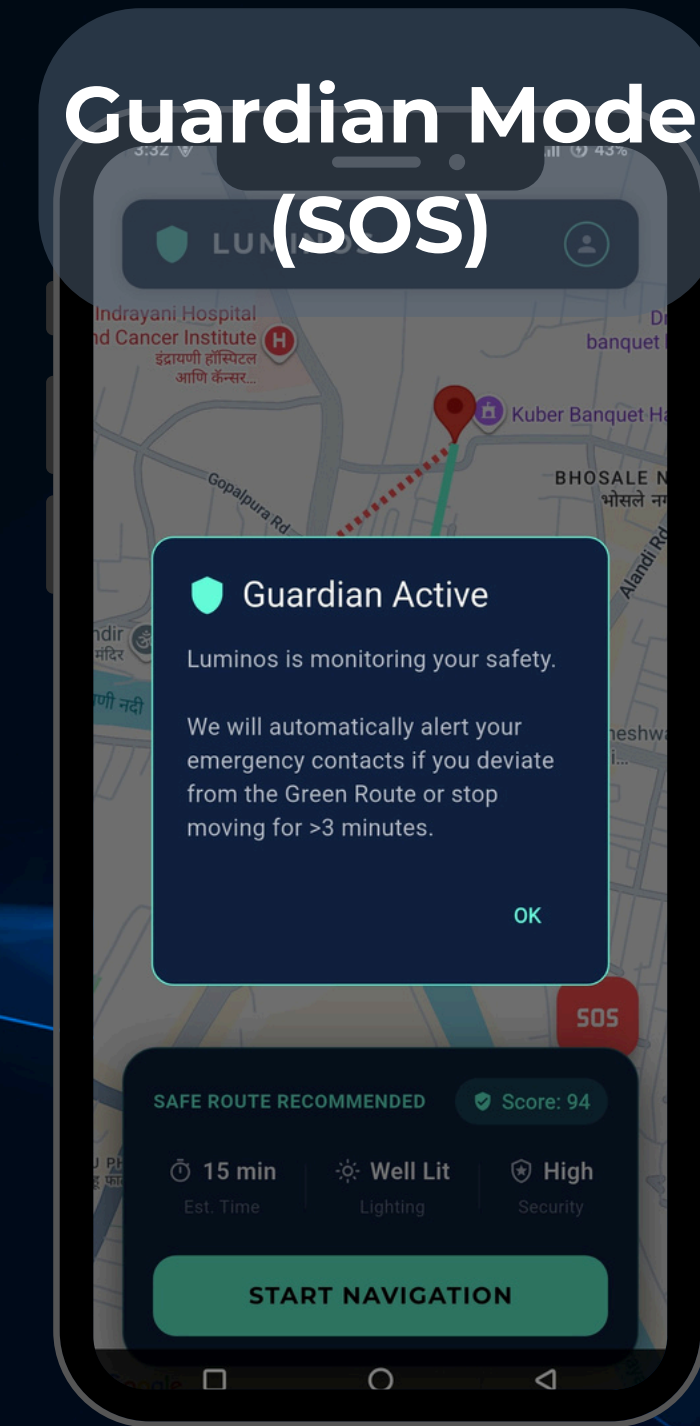
Core Capabilities



Rates every road on a scale of 1-10 based on lighting and crowd.



Green Line: Safe Route (Well-lit, Populated).
Red Line: Danger Zone (Dark, Lonely).



One-tap alert if the user deviates from the safe path or stops moving for >3 minutes.

Build with Google Technologies.

Core Google Stack



Interactive Map Interface



Fetch Path Coordinate



For identifying
shops/safe zones.

The Stack



Interactive Map Interface



For the Safety
Scoring Logic.



For storing the
simulated City Data.

From Data to Decision: The Logic Flow



User enters
Destination

App requests
routes via Google
Directions API

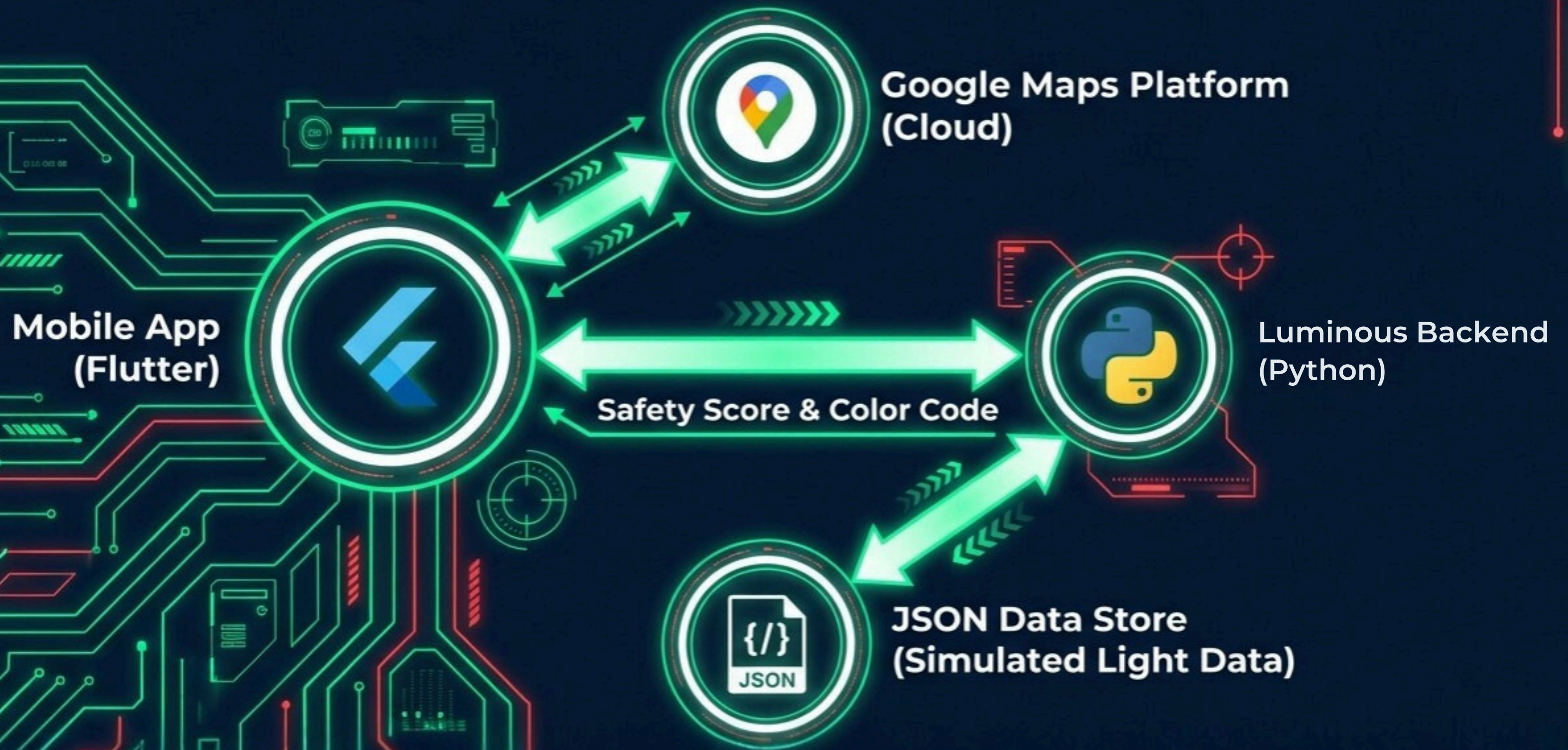
Saarthi Backend
scans route
coordinates

Algorithm checks
Lighting DB +
Places API

App highlights
Green Route on
Map

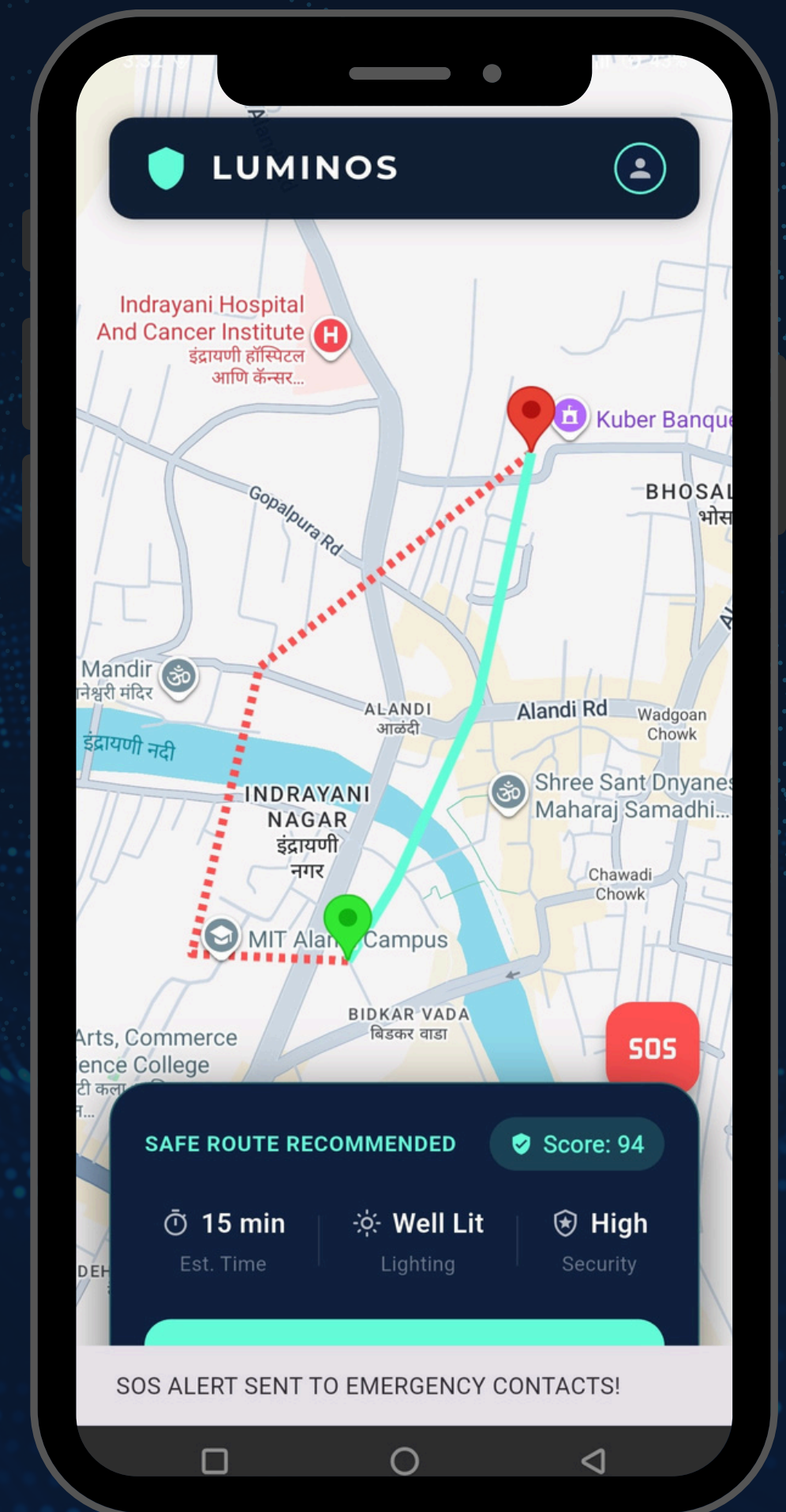
$$\text{Score} = (\text{Light} * 0.5) + (\text{Crowd} * 0.3) + (\text{Police} * 0.2)$$

System Architecture



Future Scope

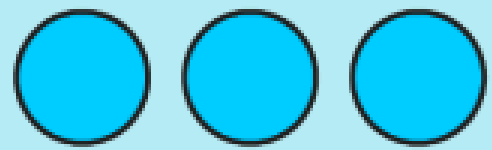
1. Crowdsourcing: Like Waze, but for safety. Users can report "Street light broken here."
2. Wearable Integration: Smartwatch vibrates if you enter a "Red Zone."
3. Government Portal: Selling this "Dark Spot Data" to the Municipal Corporation so they know where to fix lights.





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Thank you!

